



Control and Cybernetics

Feedback control and cybernetic principles laid the foundation for self-regulating systems.

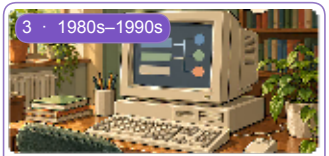
→ feedback



Planning and World Models

Planning and world models let machines reason about future states and actions.

→ future states



BDI and Agent Theory

Belief-desire-intention models formalized how agents reason and act toward goals.

→ goals & intentions



Ubiquitous and Context-Aware Computing

Computing embedded in everyday life: systems that sense context and adapt.

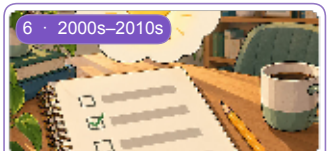
→ context



Mixed-Initiative Interaction

User input combined with agent suggestions and actions for better decisions.

→ when to step in



Goal Reasoning and Intrinsic Motivation

How agents generate goals, prioritize, and pursue what matters.

→ self-set goals



Foundation Models and Tool-Using Agents

LLMs and tools enabled flexible reasoning and execution across real-world tasks.

→ all six



Proactive AI agents

inherit all six capacities:

- Awareness
- Anticipation
- Agenda
- Arbitration
- Action
- Adaptation